

Arjuna JEE 2025

Mathematics

DPP: 6

Permutations and Combinations

- Q1** A bag contains 5 black and 6 red balls. Determine the number of ways in which 2 black and 3 red balls can be selected.
- Q2** In a cricket championship there are 36 matches. The number of teams if each play one match with other are
(A) 8 (B) 9
(C) 10 (D) 36
- Q3** In how many ways can one select a cricket team of eleven from 17 players in which only 5 persons can bowl if each cricket team of 11 must include exactly 4 bowlers?
- Q4** An automobile dealer provides motor cycles and scooters in three body patterns and 4 different colours each. The number of choices open to a customer is
(A) 4C_3
(B) 4P_3
(C) 4×3
(D) $4 \times 3 \times 2$
- Q5** The number of selecting at least 4 candidates from 8 candidates is
(A) 270 (B) 70
(C) 163 (D) none
- Q6** In how many ways can a student choose a programme of 5 courses if 9 courses are available and 2 specific courses are compulsory for every student?

Q7

A student is allowed to select at most n books from a collection of $(2n + 1)$ books. If the total number of ways in which he can select at least one book is 63, then n is

- (A) 6 (B) 3
(C) 4 (D) 5

- Q8** Suppose that 20 pillars of the same height have been erected along the boundary of a circular stadium. If the top of each pillar has been connected by beams with the top of all its non-adjacent pillars, then the total number of beams is:

- (A) 180 (B) 210
(C) 170 (D) 190

- Q9** The number of ways that a volley ball team of 6 can be selected out of 10 players so that 2 particular players are included is

- (A) 72 (B) 70
(C) 68 (D) 66

- Q10** There are 5 students in class 10, 6 students in class 11 and 8 students in class 12. If the number of ways, in which 10 students can be selected from them so as to include at least 2 students from each class and at most 5 students from the total 11 students of class 10 and 11 is $100k$, then k is equal to

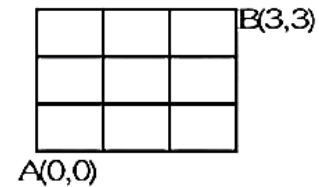
- (A) 238 (B) 17
(C) 15 (D) 14

- Q11** ${}^nC_k + \sum_{j=0}^m {}^{n+j}C_{k-1}$ is equal to
(A) ${}^{n+m+1}C_k$



- (B) ${}^{m+n}C_k$
 (C) ${}^{n+m-1}C_k$
 (D) ${}^{m+n}C_{k-1}$

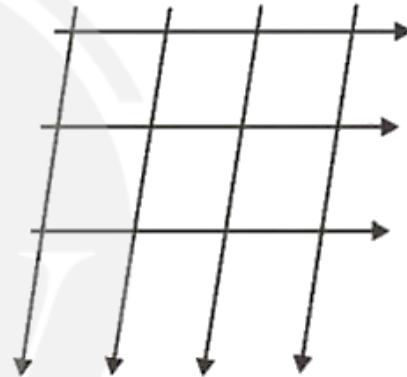
- Q12** There are n points in a plane no three of which are in the same straight line except m points which are collinear. The number of straight lines which result from joining them is
 (A) $\frac{n^2-m^2-n+m+2}{2}$
 (B) $n^2 - m^2 - n + m$
 (C) $m^2 - n^2 - n + m$
 (D) $\frac{m^2-n^2-n+m}{2}$
- Q13** The number of lines drawn through 6 points lying on a circle, is
 (A) 12 (B) 15
 (C) 24 (D) 30
- Q14** There are 15 players in a cricket team, out of which 6 are bowlers, 7 are batsman and 2 are wicketkeepers. The number of ways, a team of 11 players be selected from them so as to include at least 4 bowlers, 5 batsman and 1 wicketkeeper, is.
 (A) 139 (B) 777
 (C) 349 (D) 201
- Q15** Out of 18 points in a plane no three are in the same straight line except five points which are collinear. The number of straight lines that can be formed joining them, is
 (A) 143 (B) 144
 (C) 153 (D) None of these
- Q16** In how many ways can a person go from point A to point B if he can travel only the right or upward along the lines (Grid Problem)?



- (A) 15 (B) 18
 (C) 20 (D) 22

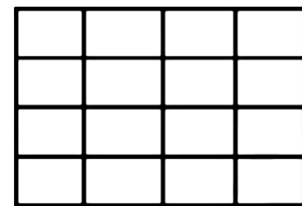
- Q17** Given five different green dyes, four different blue dyes and three different red dyes, how many combination of dyes can be chosen taking at least one green, one blue dye?
 (A) 2880 (B) 3200
 (C) 3630 (D) 3720

- Q18** The number of parallelograms in given figure, is



- (A) 6 (B) 18
 (C) 12 (D) 9

- Q19** The number of rectangles in the following figure is



- (A) 5×5
 (B) ${}^5P_2 \times {}^5P_2$
 (C) ${}^5C_2 \times {}^5C_2$
 (D) None of these

Q20



A parallelogram is cut by two sets of m lines parallel to its sides. The number of parallelograms thus formed is

- (A) $({}^m C_2)^2$
- (B) $({}^{m+1} C_2)^2$
- (C) $({}^{m+2} C_2)^2$
- (D) none of these



Answer Key

Q1 200
Q2 (B)
Q3 3960
Q4 (D)
Q5 (C)
Q6 (A)
Q7 (B)
Q8 (C)
Q9 (B)
Q10 (A)

Q11 (A)
Q12 (A)
Q13 (B)
Q14 (B)
Q15 (B)
Q16 (C)
Q17 (D)
Q18 (B)
Q19 (C)
Q20 (C)



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